SCIS 2.0 Reliability, Performance, and Refactored Processing

Over several releases, SuiteCommerce InStore (SCIS) has made internal changes to increase reliability and performance.

SCIS 2.0 continues this work by providing enhanced reliability, recovery, and record-handling for payment processing. These enhancements benefit both your sales associates and your customers by reducing the time needed to process transactions at the point-of-sale (POS).

Reliability

Reliability improvements include:

- Transaction recovery if SCIS fails when you're performing checkout for a customer
- Feedback on errors reported by the payment authorization process
- Application recovery for errors that occur on the payment device

Performance Improvements

We built in performance improvements in these areas:

- Adding lines to the cart during a Sale or Return/Refund/Exchange
- Performing an unvalidated return on an item in the cart
- Performing validated returns of items from an original transaction
- Reducing the waiting time from payment authorization to showing the authorization in the application
- Accepting any payment method, including EMV-enabled payment devices.
- The time required after completing a transaction until the application is ready for the next transaction

Audit Log Availability Improvements

You can use the SCIS Audit Log to track restricted operations that were performed during SCIS transactions. Restricted operations are events that require approval from authorized personnel, such as applying a discount over the designated limit. Audit logs are generated automatically whenever a restricted operation happens.

SCIS 2.0 lets you view audit logs from the associated transaction records. Previously, access to audit logs was limited to viewing audit log records, with links to the transactions.

Refactored Payment Processing

SCIS 2.0 provides enhanced payment processing at the POS and in NetSuite. Now, there's less waiting from tendering payment to having the receipt display. The shorter wait times are gained by pre-staging the required records and having most behind-the-scenes processing happen after the customer interaction is complete.

Another performance factor is that we use an on-demand transaction number instead of waiting to receive one from NetSuite. This on-demand number is generated locally and is later referenced to the NetSuite transaction number. We include the on-demand number in the bar code to assist with processing exchanges and refunds.

See SCIS 2.0 Payment Processing Comparison With SCIS 1.0 for more information about the SCIS 2.0 “payment first” method.

Records Removed From Transaction Processing

The following two records are no longer used to process transactions. Support for these records will continue for activities dependent on historical data, such as exchanges and refunds from past purchases.

- **Cash Sale** — If you have standard scripts that use the Cash Sale record, the SCIS team will modify them to use Invoice and Sales Order records. If you have custom scripts that depend on the Cash Sale record, you'll need to have those scrips updated.
- **Customer Payment** — Replaced with the Customer Deposit record.

Also see SCIS 2.0 Payment Processing Comparison With SCIS 1.0.

SCIS 2.0 Payment Processing Comparison With SCIS 1.0

Use this topic to compare payment processing differences between SuiteCommerce InStore (SCIS) 1.0 and the refactored flows enabled using SCIS 2.0.

- Single Payment
- Split Payment
- Shipping or Pickup Order Payment
- Exchange Without Additional Payment
- Exchange With Additional Payment
- Refund Payment

Single Payment

This payment type uses a single form of tender, like cash or a credit card.

In SCIS 1.0

Cash Sale record is staged in application memory. Payment for items is tendered, then Cash Sale is used to generate an Invoice which gets submitted. Then Customer Payment is created, submitted, and applied to the Invoice. Receipt is generated and displays.

For additional information about processing single payments using SCIS 1.0, see the help topic Cash Sales.

With SCIS 2.0

SCIS cart is ready with these records staged in application memory:

- Credit Memo
- Customer Deposit
- Sales Order

Begin transaction:

1. Items are ready to be tendered.
2. Payment is tendered.
3. Payment is added to Customer Deposit.

Submit transaction:

1. Customer Deposit with payment is submitted.
2. Receipt is generated and displays.

   Customer interaction is complete.

> **Note:** To prevent delaying customer, receipt generation isn't dependent on completing record submissions.

Prepare new records for next transaction:

- Credit Memo
- Customer Deposit
- Sales Order

Complete transaction in background:

1. Sales Order is used to generate an Invoice.
2. Sales Order is used to generate a Fulfillment Request and Store Pickup Fulfillment for the items.
3. Payment from existing Customer Deposit is applied to Invoice.

Split Payment

Split payments involve two or more tenders, such as cash and an EMV-enabled credit card.

In SCIS 1.0

Cash Sale record is staged in application memory. First payment for items is tendered, then Cash Sale is used to generate an Invoice which gets submitted. Then first Customer Payment is created, submitted, and applied to the Invoice.

Second payment is tendered. Second Customer Payment is created, submitted, and applied to the Invoice. Receipt is generated and displays.

For additional information about processing split payments using SCIS 1.0, see the help topic Invoices.

With SCIS 2.0

SCIS cart is ready with these records staged in application memory:

- Credit Memo
- Customer Deposit
- Sales Order

Begin transaction:

1. Items are ready to be tendered.
2. First payment is tendered.
3. Payment is added to Customer Deposit.
4. Second payment is tendered.
5. Payment is added to the Customer Deposit.

   A new Customer Deposit is created for each partial payment. The process continues until total payments equal amount due.

Submit transaction:

1. Customer Deposit with payments is submitted.

2. Receipt is generated and displays.

   Customer interaction is complete.

> **Note:** To prevent delaying customer, receipt generation isn't dependent on completing record submissions.

Prepare new records for next transaction:

- Credit Memo
- Customer Deposit
- Sales Order

Complete transaction in background:

1. Sales Order is used to generate an Invoice.
2. Sales Order is used to generate a Fulfillment Request and Store Pickup Fulfillment for the items.
3. Payment from existing Customer Deposit is applied to Invoice.

Shipping or Pickup Order Payment

This payment type covers transactions where a purchase is shipped to the customer or delivered to the store and held for customer pickup.

In SCIS 1.0

Cash Sale record is staged in application memory. Payment is tendered and applied to Cash Sale. Sales Order is manually created using information from Cash Sale record. This Cash Sale record is then discarded. Sales Order is billed using:

- New Cash Sale if tendered in full as a single payment
- Invoice if tendered using multiple payments

Receipt is generated and displays.

For additional information about processing shipping or pickup order payments using SCIS 1.0, see the help topic Orders for Delivery.

With SCIS 2.0

SCIS cart is ready with these records staged in application memory:

- Credit Memo
- Customer Deposit
- Sales Order

Begin transaction:

1. Items are ready to be tendered.
2. Payment is tendered.
3. Payment is added to Customer Deposit.

   If it's a split-payment, a new Customer Deposit is created for each payment. This process continues until total payments equal amount due.

Submit transaction:

1. Customer Deposit with payment is submitted.
2. Receipt is generated and displays.

   Customer interaction is complete.

> **Note:** To prevent delaying customer, receipt generation isn't dependent on completing record submissions.

Prepare new records for next transaction:

- Credit Memo
- Customer Deposit
- Sales Order

Complete transaction in background:

1. Sales Order with the items is submitted.
2. Sales Order is used to generate an Invoice.
   - If Invoice in Advance of Fulfillment is enabled, the Invoice is created and payment is applied to the Invoice.
   - If Invoice in Advance of Fulfillment is disabled, Customer Deposit is applied to the Sales Order.
3. Payment from existing Customer Deposit is applied to Invoice or Sales Order, based on:
   - If Invoice in Advance of Fulfillment is enabled, payment from existing Customer Deposit is applied to Invoice.
   - If Invoice in Advance of Fulfillment is disabled, payment from existing Customer Deposit is applied to Sales Order.
4. Fulfillment process occurs:
   - If items are marked to be shipped, a fulfillment request is created.
   - If items are marked to be picked up on location (cash and carry), the order is automatically fulfilled.

Exchange Without Additional Payment

Rather than receive a cash or credit card refund, a customer can request an item exchange. This payment type covers transactions where the item being returned and the item received in exchange are the same price.

> **Tip:** With the exception of validating the return, the payment flows for validated and unvalidated exchanges are the same.

In SCIS 1.0

SCIS cart is ready with Cash Sale record staged in application memory. Return is validated and tendered, then Cash Sale is used to generate an Invoice, which gets submitted. Return items are added to a Credit Memo, with the memo applied to an Invoice and submitted. Receipt is generated and displays.

For additional information about processing exchanges without an additional payment using SCIS 1.0, see the help topic Refund by Exchange.

With SCIS 2.0

SCIS cart is ready with these records staged in application memory:

- Credit Memo
- Customer Deposit
- Sales Order

Begin transaction:

1. Return is validated and items are ready to be tendered.
2. Return items are added to Credit Memo.
3. Exchange items are added to Sales Order.
4. Payment of $0.00 is tendered.

Submit transaction:

1. Sales Order is submitted.
2. Credit Memo is submitted.
3. Receipt is generated and displays.

   Customer interaction is complete.

> **Note:** To prevent delaying customer, receipt generation isn't dependent on completing record submissions.

Prepare new records for next transaction:

- Credit Memo
- Customer Deposit
- Sales Order

Complete transaction in background:

1. New Sales Order is created for next transaction.
2. Sales Order is used to generate an Invoice.
3. Credit Memo is applied to the Invoice.

Exchange With Additional Payment

This type of exchange happens when the item being returned is priced less than the item received in exchange. In this scenario, the customer must tender payment to cover the price and tax difference.

> **Tip:** With the exception of validating the return, the payment flows for validated and unvalidated exchanges are the same.

In SCIS 1.0

SCIS cart is ready with Cash Sale record staged in application memory. Return is validated and payment is tendered. Cash Sale is used to generate an Invoice, which gets submitted.

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Return items are then added to a Credit Memo, with the memo applied to the Invoice and submitted. Customer Payment is created, submitted and applied to the Invoice. Receipt is generated and displays.

For additional information about processing exchanges with an additional payment using SCIS 1.0, see the help topic Refund by Exchange.

With SCIS 2.0

SCIS cart is ready with these records staged in application memory:

- Credit Memo
- Customer Deposit
- Sales Order

Begin transaction:

1. Return is validated and items are ready to be tendered.
2. Return items are added to Credit Memo.
3. Exchange items are added to Sales Order.
4. Payment equal to the difference of return items and exchange items is tendered.

Submit transaction:

1. Sales Order is submitted.
2. Credit Memo is submitted.
3. Customer Deposit with payment is submitted.
4. Receipt is generated and displays.

   Customer interaction is complete.

> **Note:** To prevent delaying customer, receipt generation isn't dependent on completing record submissions.

Prepare new records for next transaction:

- Credit Memo
- Customer Deposit
- Sales Order

Complete transaction in background:

1. Sales Order is used to generate an Invoice.
2. Credit Memo is applied to the Invoice.
3. Payment from existing Customer Deposit is applied to Invoice.

Refund Payment

A refund is given when a customer returns an item without receiving another item in exchange. The customer gets back the amount that they paid for the returned item.

Tip: With the exception of validating the return, the payment flows for validated and unvalidated refunds are the same.

In SCIS 1.0

SCIS cart is ready with Cash Sale record staged in application memory. Return is validated and refund is tendered. Cash Sale is used to generate an Invoice, which gets submitted. Return items are then added to a Credit Memo, with the memo applied to the Invoice and submitted.

Credit Memo is applied to a Customer Refund. Customer is provided the refund amount. Receipt is generated and displays.

For additional information about processing refund payments using SCIS 1.0, see the help topic Validated Returns.

With SCIS 2.0

SCIS cart is ready with these records staged in application memory:

- Credit Memo
- Customer Deposit
- Sales Order

Begin transaction:

1. Return is validated and items are ready to be tendered.
2. Refund payment is tendered.

Submit and complete transaction:

1. Return Authorization is created.
2. Return items are added to Return Authorization.
3. Return Authorization is used to populate Credit Memo.
4. Credit Memo is submitted.
5. Customer Refund is created and submitted and linked to the Credit Memo.
6. Receipt is generated and displays.

   Customer interaction is complete.

Note: To prevent delaying customer, receipt generation isn't dependent on completing record submissions.

Hold unused records for next transaction:

- Customer Deposit
- Sales Order

Prepare new record for next transaction:

- Credit Memo